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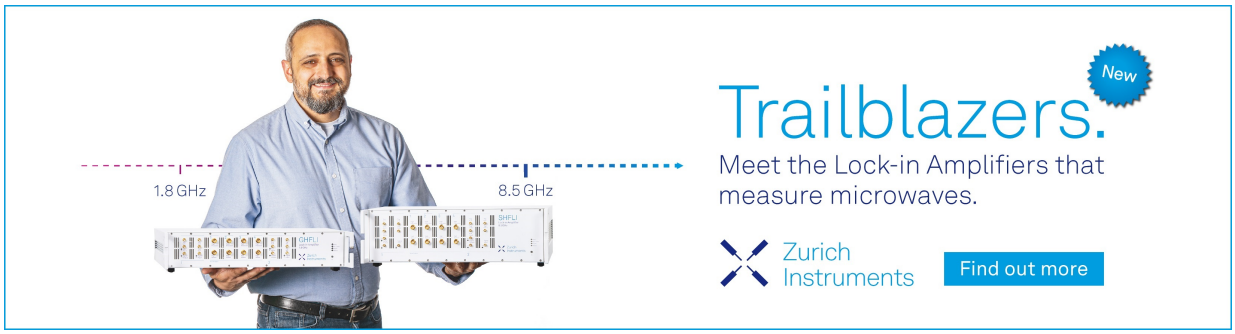

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Human Face Detection in a Crowd Image Based on Template Matching Technique

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Abstract. In numerous security applications, face identification and recognition are astounding and significant capacities that we use in our daily lives. The primary purpose behind the interest in building up the computer vision-based robotized advances of mechanized face acknowledgment emerges from the genuine worries for public security in the present organized world. Where personality check's for physical and coherent access in numerous offices are basic in daily life.The process of identifying a human face image through a crowd is an important and very useful task for security purposes. Usually, digital cameras are erected on the roofs of dwellings, buildings, and storefronts to monitor events and accidents on the opposite streets. Often, the captured photos involve a group of people who are at that location. In the event of a strange accident, security services are required to identify the perpetrator. If these security services have pictures of suspects, then we can compare and match their photos with the captured image to identify the real perpetrator. Traditionally, It includes three processing steps in a crowd analysis.That is, preprocessing, object detection, and event/behavior recognitionIn this paper, we will try to recognize a person from a crowd of people by using a single image of the person to be recognized, using the Two-Dimensional Normalized cross-correlation function.

Keywords: Face Identification, Facial recognition, Normalized Cross-correlation, Template matching

INTRODUCTION

Among all the biometric techniques for human recognition and human verification, fingerprint and face recognition have been utilized all the more as often as possible its materialness is simpler, and the working reach is more extensive than different systems like; iris examining, signature, and so forth. Fingerprint Detection is extremely old and has been utilized for a long, however, face recognition has gotten better known for the last a quarter-century. It is essential for facial picture handling applications and it is one of the difficult subjects in IT research. Finding any face in a given picture is face recognition [1]. Face detection is an important initial phase in face "recognition systems", to limit and removing the face district from the background. It likewise has a few applications in territories like substance-based picture recovery, video coding, video conferencing, swarm observation, and intelligent human-computer interfaces [2].

Automatic face detection has numerous business and security applications in character approval and recognition and has gotten perhaps the most blazing subject nearby image processing and design recognition since 1990. The accessibility of achievable advances just as the expanding demand for reliable security systems in this day and age has been an inspiration for some analysts to grow new techniques for face recognition. Programmed acknowledgment of human faces keeps on drawing in analysts from different areas like computer vision, picture handling, design acknowledgment, neural networks, and psychology. Furthermore, has been progressively acknowledged by the overall population to be utilized in authentication, security, and law implementation, [3]. This work will explicitly contemplate human bunches, as animal categories that is typically the most significant from a commonsense perspective. A bunch can be characterized as a gathering of items spatially near a specific class. There

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are a few reasons why swarm location is a test. In the first place, the restricted goal of the pictures implies that proof for a solitary individual is fairly scant. Given that incomplete blockages are plentiful in swarms and the assortment in dress, stance, and lighting, instead of extraordinary openness of people as a key segment. In development, it's anything but a promising methodology [4,5].

FACE RECOGNITION

Face Recognition is essential for a bigger setting considered biometrics that gives us the idea of life measure; it very well may be characterized as the physiologic or mental attributes that can be utilized to check a person's personality. Three fundamental steps are used to build up a strong face recognition system as demonstrated in fig.(1),.*Face identification* is the initial step used to recognize and find the human face image got by the system. *Feature extraction* is the subsequent advance utilized to extricate the element vectors for any human face situated in the initial step. Ultimately, *the face recognition* step incorporates the features extracted from the human face to match it and all layout face data sets to choose the person face character [6] .

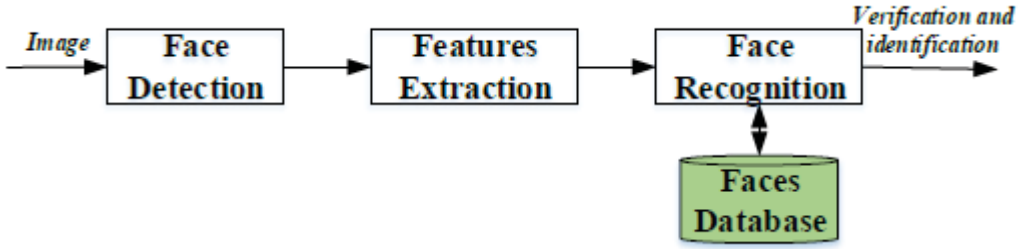

FIGURE 1, Essential Steps of Face Recognition Systems [6].

Biometrics procedures are isolated into two classifications: physiologic and mental. Physiologic biometrics is based only on the physiologic aspects of the body, and the most common are the face, fingerprint, hand geometry, iris, and retina. . Mental biometrics depends predominantly on mental angles, and the most widely recognized are written by hand mark, and voice. The biometrics framework is basically an example acknowledgment framework that makes an individual recognizable proof deciding the validness of a particular physiologic or mental trait of a person. The biometrics frameworks are exposed to the Guideline of Limit. Appropriately, a face is perceived if its highlights lie inside on the recognized range [7,8]. This standard characterizes some vulnerability degree in outcomes that infer getting more than one answer as a looking through outcome, in the end requiring human mediation for the right elective decision. In light of a mental perspective, there are two face acknowledgment levels: entrance level and subordinate level. At the passage level, every one of the appearances is seen as a lone face classification, and in the subordinate level, the individual countenances are recognized by more point by point attributes [9].

FACE DETECTION METHODS

Face detection methods are classified into four classes and they are *feature-based class, Recognition-based class, template-based class and appearance-based class*[10]..

- Feature based techniques utilize the special compoents of the face. Components like eyes, nose, and lips are utilized to distinguish a face. The construction and size of these components are not quite the same as the nonface object. These highlights might be introduced by sections, borders, and territories of certain figures shaped by the identified focuses. The distinguished (for example distances and positions) can be utilized to handle the face cycle activity .
- Recognition -based class these standard-based techniques represent human information on what comprises a run-of-the-mill face. Normally, the principles catch the connections between facial highlights; these techniques are fundamentally intended for facial localization.
- Template-based class is done a few standard examples of a face are put away to portray the face all in the entire facial highlights independently. The connections between's an information picture and the put-away examples are processed for identification. These strategies have been utilized for both face confinement and recognition.

- Appearance-based class, rather than template matching, the models or formats are gained from a bunch of preparing pictures that should catch the agent fluctuation of facial appearance. These learned models are then utilized for location. These methods are arranged on a very basic level for face recognition [11].

FACE IDENTIFICATION AND VERIFICATION

Biometric methods are utilized for two capacities, identification or verification. The determination and execution of the innovation are firmly identified with this goal. In recognizable proof, the objective is to distinguish an individual dependent on a correlation of highlights gathered against a data set of recently gathered examples [12]. In other words, systems that are intended with the end goal of recognizable applications will address the inquiry: "Who the subject is?" .In the check applications, then again, we want to confirm whether the subject is the individual that they guarantee to be. This is finished by approving the gathered highlights against a formerly gathered component test for the person in our library. Biometric methods designed for verification purposes answer the inquiry: "Is the subject who he says he is?"

RELATED WORK

This section will highlight the literary survey to get a general idea about the research. The present survey includes significant previous works related to detecting the image of a face person based on pattern recognition techniques. Several research papers have been published in the area of face detection. Selected research works will be discussed in this section for understanding the issues and techniques used for handling the issues. There are many writers who describe several techniques for detecting the face. Some of these methods will be reviewed in the following:

Eigenface; is one of the popular technique for holistic (or subspace) approaches used to extract features points of the face image. It is perhaps the most completely researched way to deal with face recognition. It is otherwise called Karhunen Loève (KT) development, Eigen pictures, eigenvector, and principal component analysis (PCA). Since it depends on (PCA) technique. *M. Kirby and L. Sirovich* [13] utilized head part investigation to effectively address pictures of appearances, as outlined in fig. (2), they contended that any face pictures could be roughly remade by a little assortment of loads for each face and a standard face image. The loads portraying each face are acquired by projecting the face picture onto the Eigen image. The weights describing each face are obtained by projecting the face image onto the eigenpicture.

Ibnu Utomo M. et al. [14], in this research, investigated the efficiency of the PCA Eigenface system in recognizing human face images from multiple databases, each with its own sets of difficulties, such as lack of illumination of facial images, major variations in speech, and the use of accessories like glasses. The identifiable accuracy varies widely, ranging from 100% to 67% in each database database with and with an average recognition of more than 85%.

FIGURE 2. Examples of some eigenfaces[13].

Neural Networks, the attractiveness of using artificial neural networks (ANN) could be due to its nonlinearity in the network. Hence, the feature extraction step by using (ANN) techniques may be more efficient than the linear

(Karhunen-Loève) methods. The Single Layer Adaptive Network Wilkes, Stonhan and Aleksander Recognition Device (WiSARD) was one of the first techniques for Artificial Neural Network (ANN) to contain a separate network for each stored individual [15]. One of the most important determinants of successful recognition is the choice of the method in which the neural network is constructed. It largely depends on the intended application. To reveal the multi-layered face receptor, a convolutional artificial neural network (CANN) was applied. *Wankou Yang, et. al*. [16] they proposed a new cascaded convolutional neural network method consisting of two main stages. During the first stage, low-pixel candidate window is used as an input such that the shallow convolutional neural network quickly extracts the candidate window. In the second stage, the window from the former stage is resized and used as an input to the corresponding network layer respectively. The whole network achieves improved performance on the FDDB and PASCAL face datasets.

Template Matching, A simple variant of template matching is to look at a test picture as a 2D collection of force values using a logical measurement, such as "the Euclidean distance," with a single layout addressing the entire face. It's a computer vision technique for locating a subimage of an objective image that corresponds to a template image. This technique is commonly used in ob. The most important thing is to use a proper "measure" to assess similitude or coordination. At the pixel level, as well as at a higher level, template matching has been done. Complete templates, partial templates, piece templates, and flexible templates are the four forms of pixel templates. High-level block-matching approaches, on the other hand, operate on a picture that has been divided into areas of interest. There are a few other more modern face detection adaptations of template matching. To address a person's face, one can use multiple face formats from different perspectives. A single perspective of a face can be addressed by a variety of different, particularly more modest formats [18,19]. Prior to coordinating, the face image of dark levels can also be properly treated.

***Line Edge Map Recognition Method*:** Edge info is a valuable component for addressing objects that are coldhearted toward lighting changes partly. (LEM) is generally utilized in different fields of example acknowledgment, see Fig. (3). This approach, as proposed by Gao and Leung [20], this methodology can be considered as a mix of format coordinating and mathematical component coordinating. They showed that it is achievable to see faces using an edge-line map (LEM). A comprehensive assessment of the proposed thought is driven which covers all pieces of human face affirmation, for instance, face affirmation under; controlled/ideal conditions and size assortments, varying lighting conditions, changing outward appearances, and moving stances. This assessment shows that the LEM, alongside the proposed nonexclusive line-area Hausdorff distance measure, gives another strategy to go up against coding and affirmation .There are many other methods have been proposed to recognize faces, and there is no room for them to be mentioned and mentioned in our current research. For viewing, one can return to [21, 22].

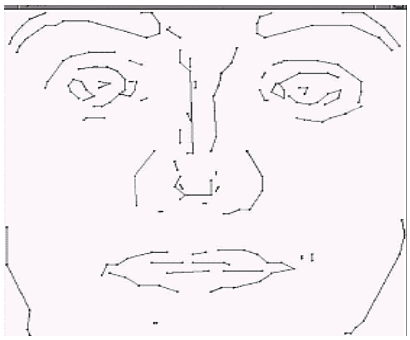

FIGURE 3. An illustration of a face LEM. [20]

FACE DETECTION RESEARCHES

Face recognition techniques (heuristic and anthropometric) have been used since the early 1970s [23]. Due to various assumptions such as a blurred image, a front-face scenario, and a standard passport picture, these technologies are mostly rigid. Any alteration in image conditions necessitates fine tuning, if not a complete overhaul, for these systems. Despite these issues, there has been a surge of interest in redevelopment, [24] . Face detection is a difficult problem in computer vision since the human face is a complex entity with a high degree of

variability in appearance. There have been a range of techniques suggested, ranging from basic edge-based algorithms to composite high-level approaches using sophisticated pattern recognition methods,[25].

One of the most common problems facing automatic face detection is that the image must be the face facing the camera and include all the details (such as: eyes, nose and mouth). Therefore, in many cases, we resort to manually identifying the faces which are included in the image. A manually identifying faces by creating sharp corners rectangles, with desired size surrounded them.

TEMPLATE MATCHING BY CROSS-CORRELATION

An important procedure in face recognition is the face-matching method. This research proposes a face-matching method to recognize a person from a crowd of people using a single image of the person and begin to search for matches with the different images of the same person using the Two-Dimensional Normalized Cross-correlation function. The motivation behind the use of cross-correlation to match the template is by distance measurement (square of Euclidean distance) [26].

$$d^2_{f,t}(u,v) = \sum_{x,y}\left[f(x,y) - t(x-u, y-v)\right]^2 \text{.....(1)}$$

Where *f* is the image and the sum is over *(x,y)* under the window containing the feature *t* positioned at *(u, v)*. In the expansion of $d^{2.}$

$$d^2_{f,t}(u,v) = \sum_{x,y}\left[f^2(x,y) - 2f(x,y)t(x-u, y-v) + t^2(x-u, y-v.)\right] \text{..... ..(2)}$$

If the terms ($\sum_{x,y} t^2(x-u, y-v)$ and $\sum_{x,y} f^2(x,y)$) are approximately constant then the remaining Cross-Correlation (CC) term ;

$$c(u,v) = \sum_{x,y} f(x,y)t(x-u, y-v) \text{..... ...(3)}$$

Cross-correlation
term is a measure of the similarity between the image and the feature. There are several disadvantages to using (*equation 3*) for template matching:

- If the image energy $\sum_{x,y} f^2(x,y)$) varies with position, matching using (*equation* 1) can fail.

For example, the correlation between a feature and a bright spot could be more than the correlation between that feature and the exact matched area of the image.

- The size of the function determines the range of c (u, v).
- Equation (3) is not constant for changes in the image amplitude such as those caused by changing lighting conditions in the image.

By normalizing the image and the feature vectors to unit length, resulting in a coefficient of correlation similar to a cosine; The NCC formula is defined as follows:

$$\gamma(u,v) = \frac{\sum_{x,y}\left[f(x,y) - \bar{f}_{u,v}\right]\left[t(x-u, y-v) - \bar{t}\right]}{\left[\sum_{x,y}\left[f(x,y) - \bar{f}_{u,v}\right]^2 \sum_{x,y}\left[t(x-u, y-v) - \bar{t}\right]^2\right]^{1/2}} \text{..... ...(4)}$$

.

Where ($\tilde{t}$) is the mean of the feature and ($\widetilde{f_{u,v}}$) is the mean of *f(x, y)* in the region under the feature. Refer to $\gamma(u,v)$ as ***normalized cross-correlation*** (NCC). Here *(x, y)* and *(u, v)* represent pixel coordinates.

The main reasons behind using NCC are:

• NCC remains unchanged upon changing the global brightness that means image intensity differences have a much lesser effect on matric correlation

• The correlation value between two images varies from [-1, +1], while for a normal correlation in it only tends to +1, which is easy to understand.

METHODOLOGY

In the proposed method, a template identical to the target image is created with or without the target image scanned by the template image. Our methodology summarized in below steps.

Step 1*: Input suitable size and number of select face on the original crowed image;*

Step 2: *Double click on the middle points of each face to identifying the faces in the crowed image;*
Step 3: *Manually selecting part of the single image (the template image) to be recognized;*
Step 4: *Convert the target and template image into gray scale images;*
Step 5: *Matching the single image of step (3) with the crowd images of step (2) using the 2-D Normalized Cross-Correlatio (NCC) to determine the maximum correlation;*
Step6*: Templet image found in original image.*

EXPERIMENTAL RESULTS

The written MATLAB program, in our present research, has been run by two ways;
1.If the single image (the Template to search for) was part of the crowd image; The program in this case run directly to match the smaller size (Template) image with the larger size (crowed) image. Fig. (4) illustrates the output matching and cropping result.

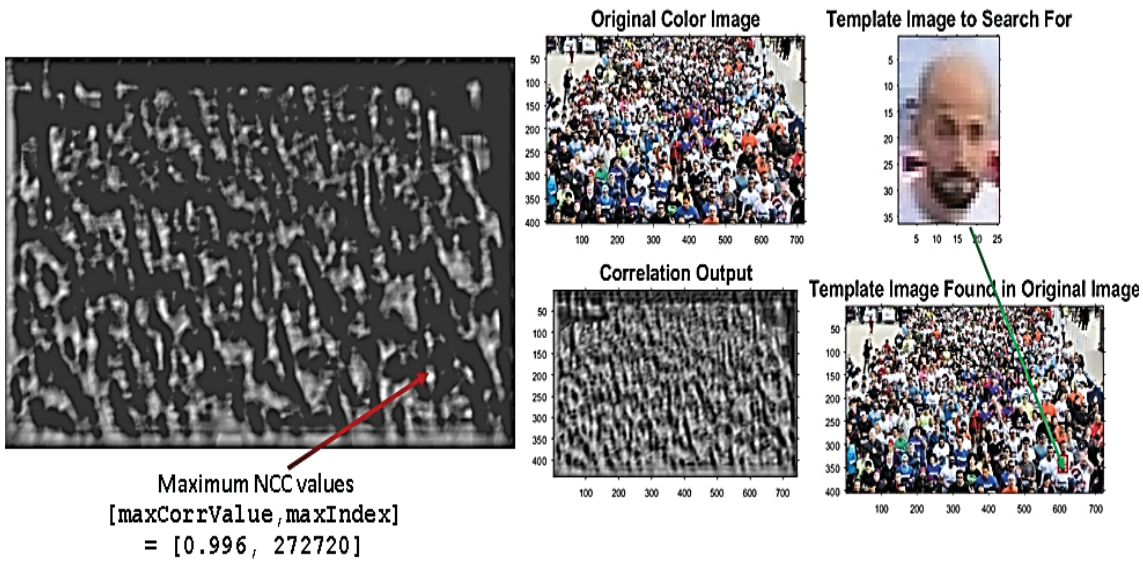

FIGURE 4, the identification process by adopting the maximum NCC value.

Matching single differently captured (Template to search for) image with manually selected faces: The mentioned in the step (4) of the methodology part of the single image to be recognized should selected manually with the assurance that it includes the important details found in the crowd image.The Normalized Cross-Correlation (NCC) can then implement to locate the position of the differently captured (single) image in the crowed image.

CONCLUSION

A very encouraging matching, verifying, and recognition results have been satisfied by implementing the normalized-cross-correlation (NCC) algorithm. The result can be improved if pure face images were adopted (i.e. background-free), and having the same scales (i.e. viewed by the same camera at a fixed distance from the person's face, and using unvaried illumination). However, certain image preprocessing operations may still require enhancing

the results (e.g. image normalization, image re-centering, etc.). Moreover, the size selection of the rectangle surrounding the faces (in the second recognition method) of the crowd image should be compatible with the spatial resolution of the included faces within the original crowed image.

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